Machine Learning-Based Prediction of Semiconductor Band Gap

Physics-Informed Materials Descriptors and 3D structure analysis

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Presentation Outline

1. Background
2. Dataset and Features
3. Machine Learning Methodology
4. Results
5. Materials Analysis
6. Discussion and Conclusion

Understanding the Semiconductor Band Gap

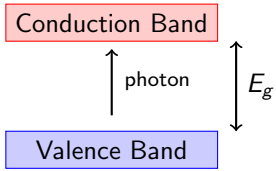
What is the Band Gap?

The band gap E_g is the energy separation between the valence-band maximum and conduction-band minimum.

- It strongly influences electrical transport and optical absorption.
- A photon can promote an electron across the gap when

$$h\nu \geq E_g.$$

- Band-gap engineering is important for photovoltaics, LEDs, photodetectors and other optoelectronic devices.



Electronic excitation across the band gap

Key Relationship

$$\boxed{h\nu \geq E_g}$$

Photon energy must be sufficient to excite an electron across the band gap.

Why Predict Band Gap from Composition?

- Experimental band-gap determination can require synthesis and characterization.
- Chemical composition contains information about atomic mass, atomic number and electronegativity.
- Machine learning can learn nonlinear relationships between composition-derived descriptors and measured band gaps.
- The project therefore connects **materials physics + descriptors + machine learning**.

Band Gap and Optical Absorption

Absorption-Edge Relation

For a simplified absorption-edge estimate:

$$\lambda_{\text{edge}} \approx \frac{1240}{E_g} \text{ nm},$$

where E_g is the band gap in eV.

- Larger $E_g \Rightarrow$ shorter-wavelength edge.
- Smaller $E_g \Rightarrow$ longer-wavelength edge.

Example

A predicted band gap of

$$E_g = 2.570 \text{ eV}$$

gives

$$\lambda_{\text{edge}} \approx \frac{1240}{2.570} = 482.490 \text{ nm}.$$

This wavelength lies in the **visible region**.

Important Note

This is an approximate conversion and not a complete optical-property model.

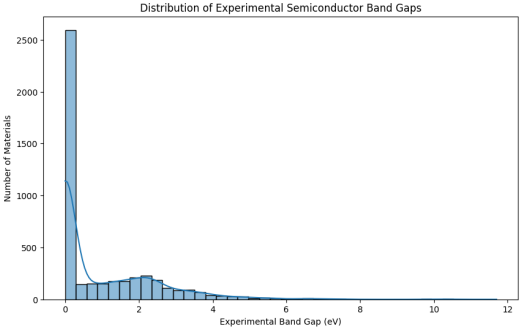
Dataset Used in the Project

Dataset Overview

- **Dataset:** `matbench_expt_gap`
- **Samples:** 4,604 materials
- **Input:** Chemical composition
- **Target:** Experimental band gap (eV)

Data Preparation

4,604 materials remained after the loading and cleaning steps.



Distribution of experimental band gaps

Physics-Informed Composition Features

Structural Features

- Number of elements
- Total atoms in reduced formula

Atomic Properties

- Average atomic mass
- Maximum atomic number
- Minimum atomic number

Electronegativity Features

- Maximum electronegativity
- Minimum electronegativity
- Electronegativity difference

Why These Features?

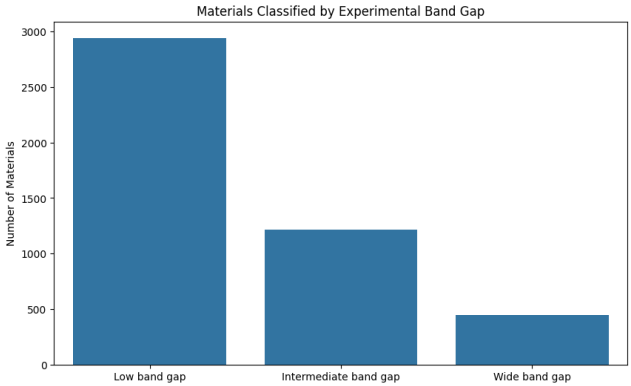
They convert chemical composition into a compact numerical representation that can be directly used by regression models.

Composition $\longrightarrow$ *Physics-informed descriptors* $\longrightarrow$ *Band-gap prediction*

Experimental Band-Gap Categories

The notebook groups materials as:

- **Low band gap:** $E_g < 1.0$ eV
- **Intermediate band gap:** $1.0 \leq E_g < 3.0$ eV
- **Wide band gap:** $E_g \geq 3.0$ eV



Models Evaluated

1. Linear Regression
2. Random Forest Regression
3. Gradient Boosting Regression
4. Deep Neural Network

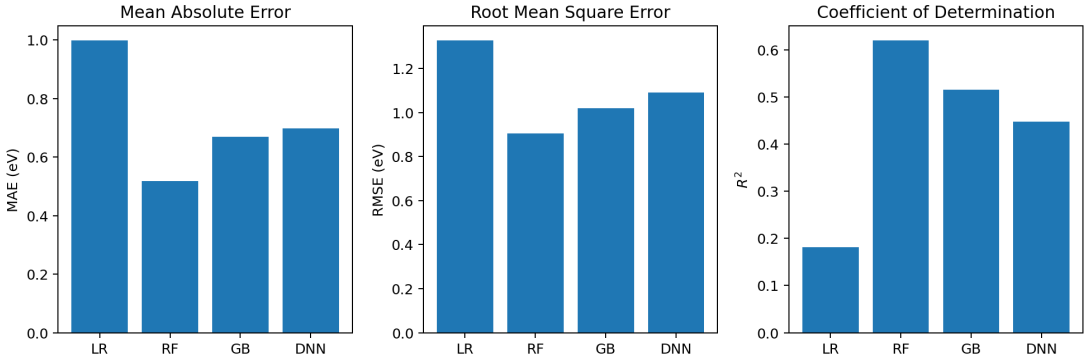
Data split

The notebook uses an 80/20 train–test split with `random_state=42`.

Deep network

The implemented network uses dense layers of 128, 64 and 32 units, batch normalization, dropout and a linear output for continuous band-gap prediction.

Model Performance

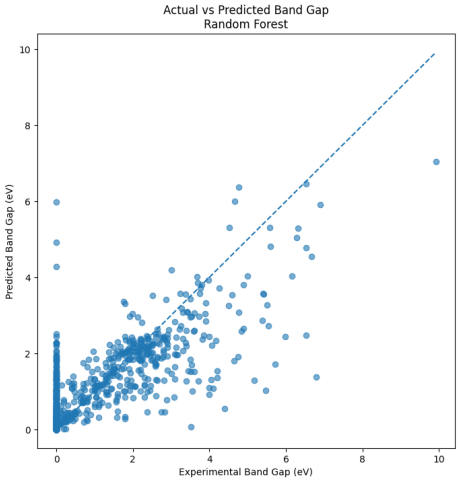


Model Performance: Numerical Results

Model	MAE (eV)	RMSE (eV)	R^2
Linear Regression	0.997824	1.326643	0.181688
Random Forest	0.518957	0.904775	0.619380
Gradient Boosting	0.669760	1.020542	0.515747
Deep Neural Network	0.698311	1.090020	0.447568

- In the notebook's MAE-based selection, Random Forest is selected as the model used for the subsequent actual-vs-predicted and error plots.

Actual vs Predicted Band Gap



Model Agreement

The plot compares the experimental band-gap values with the values predicted by the regression model.

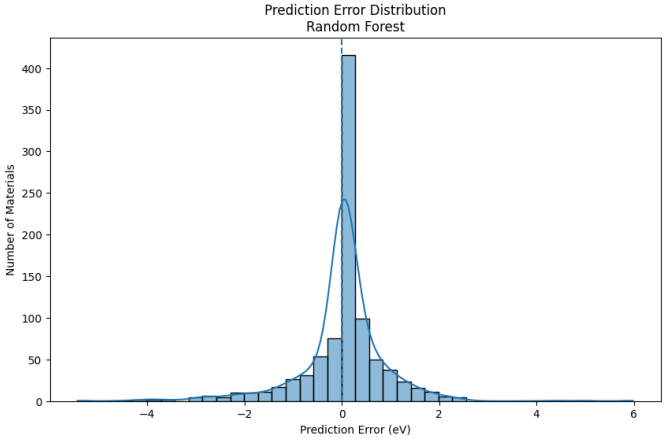
Ideal Prediction

The dashed diagonal line represents ideal agreement:

$$E_g^{\text{pred}} = E_g^{\text{actual}}.$$

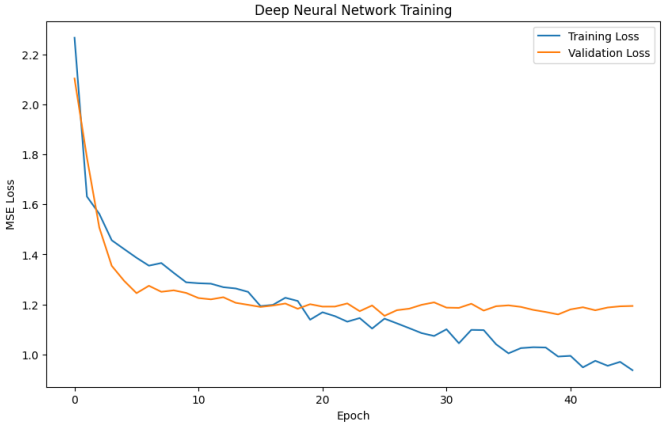
Points closer to the diagonal indicate closer agreement between experimental and predicted band-gap values.

Prediction Error Distribution



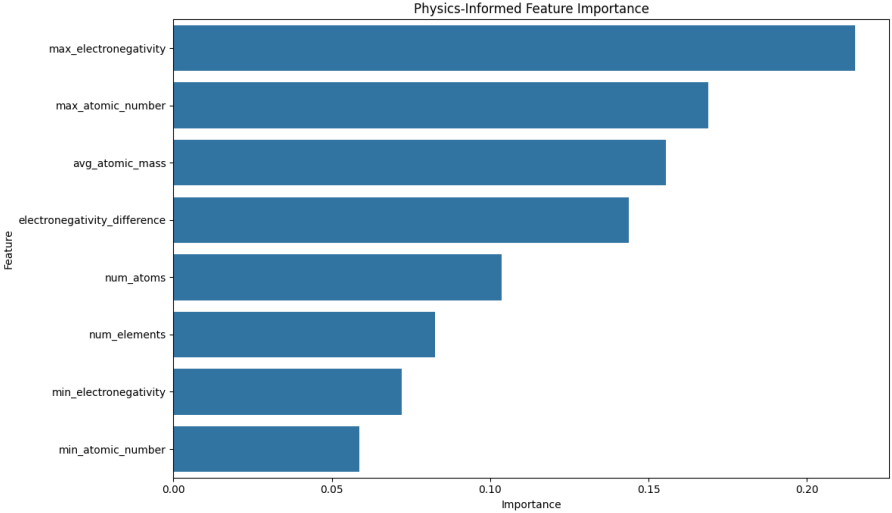
Prediction error is defined in the notebook as predicted band gap minus experimental band gap.

Deep Neural Network Training



Training and validation loss curves are used to inspect learning behavior during training.

Feature Importance



Example: TiO_2 Prediction

Recorded notebook result

Predicted band gap for TiO_2: **2.510 eV**.

A separate optical-analysis example in the notebook reports:

$$E_g = 2.570 \text{ eV}, \qquad \lambda_{\text{edge}} \approx 494 \text{ nm}.$$

- These outputs illustrate how a predicted band gap can be connected to a simple optical-edge estimate.

Example: GaAs Complete Analysis

The notebook's complete material-analysis example reports:

- Composition: GaAs
- Number of elements: 2
- Predicted band gap: 1.008 eV
- Estimated absorption edge: 873.24 nm
- Spectral region: Near Infrared (NIR)

Key Findings

- The project demonstrates an end-to-end workflow from experimental composition–band-gap data to machine-learning prediction.
- Composition-derived physics-inspired descriptors can be used as model inputs.
- Four regression approaches were evaluated.
- The recorded test-set metrics show different levels of predictive performance across the models.
- The notebook uses Random Forest for the subsequent best-model visualization based on the lowest MAE.

Live Streamlit Application: Band-Gap Prediction

Interactive Machine Learning Platform Local deployment:

`http://localhost:8501`

Conclusion

- Machine learning provides a practical route for rapid screening of materials by estimated band gap.
- The project combines fundamental semiconductor physics with materials informatics.
- The workflow can be extended toward candidate screening for photovoltaic and optoelectronic materials.

Thank You